

Exploring the Probabilistic Causal Structure of Media Trust: A Bayesian Network Analysis

Hye Hyun Ma

Ph.D. Candidate, Dept. of Advertising & Public Relations, Hanyang University

2025. 5. 31.

Background

- Trust in media and journalism has become increasingly complex in the digital era marked by algorithmic curation, information fragmentation, and rising misinformation (e.g., Fink, 2019; Jakobsson & Stiernstedt, 2023; Park et al., 2020; Tsfaty & Ariely, 2014).
- Public trust in journalism affects how people seek, interpret, and use information, and has broader implications for democracy, public discourse, and social cohesion (Hanitzsch et al., 2018; Newton, 2001).
- => (Motivation) However, most existing empirical studies rely on linear modeling, failing to capture conditional and interactive dependencies among predictors.

Conceptualizing Media Trust (Fawzi et al., 2021)

- Media trust consists of:
 - Dimensions of trust:
 - **Trust in the content:** perceived accuracy, fairness, completeness
 - **Trust in media actors:** perceived competence and integrity of journalists and institutions
 - **Audience predispositions:** cognitive, affective, and behavioral tendencies shaped by ideology, media expectations

Conceptualizing Media Trust (Fawzi et al., 2021)

- Media trust is shaped by individual-level antecedents, which include
 - **(a) Social characteristics:**
 - Interpersonal trust tends to increase media trust.
 - Misanthropy, conspiracy mentality, and hostile media perceptions correlate with lower media trust.
 - **(b) Political characteristics:**
 - Institutional trust and political efficacy are positively related to media trust.
 - Political ideology and extremity affect levels of trust: conservatives and populists often report lower trust.
 - Political interest, knowledge, and perceived efficacy correlate positively.

Conceptualizing Media Trust (Fawzi et al., 2021)

- Media trust is shaped by individual-level antecedents, which include
 - **(c) Media-related characteristics:**
 - Media literacy, perceived professionalism, and source diversity all influence trust.
 - Exposure to specific content and perceptions of media bias also play roles.
- These interact dynamically and recursively to shape levels of trust in media.

Conceptualizing Media Trust (Fawzi et al., 2021)

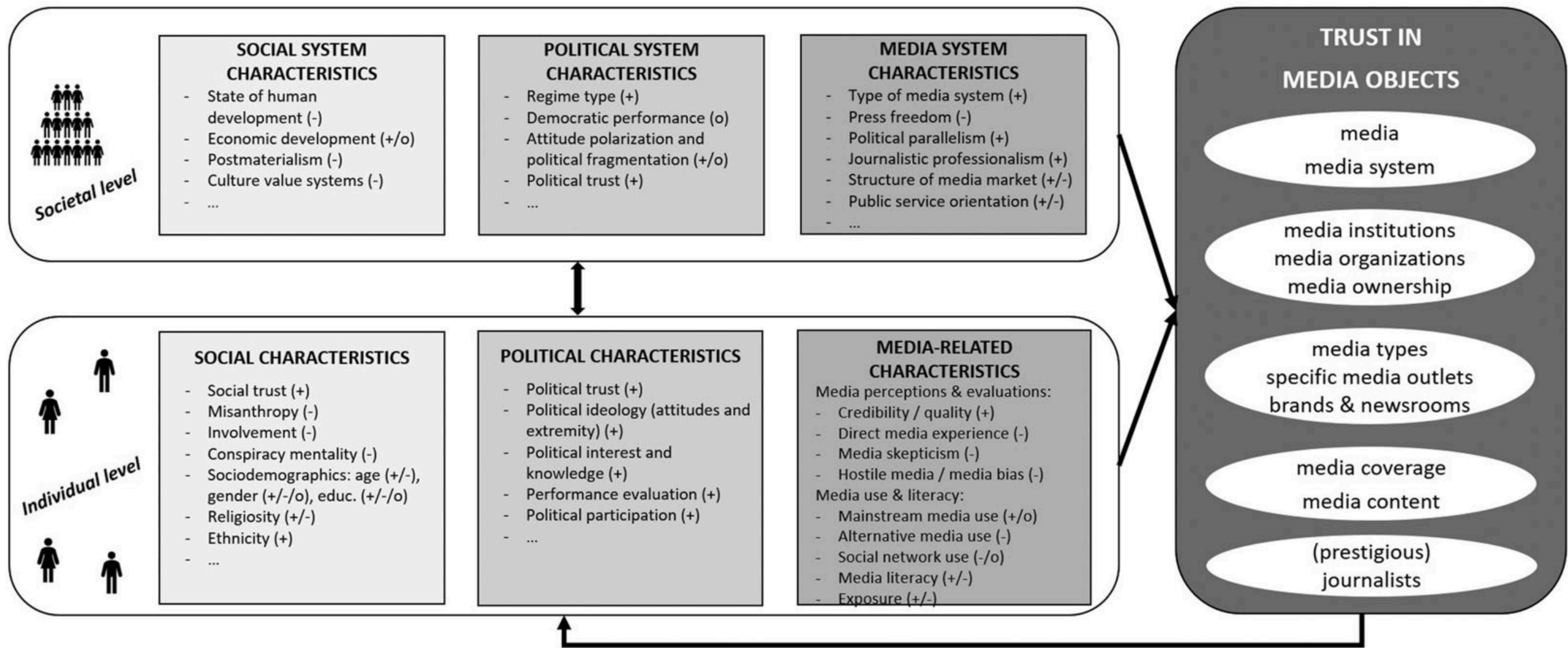
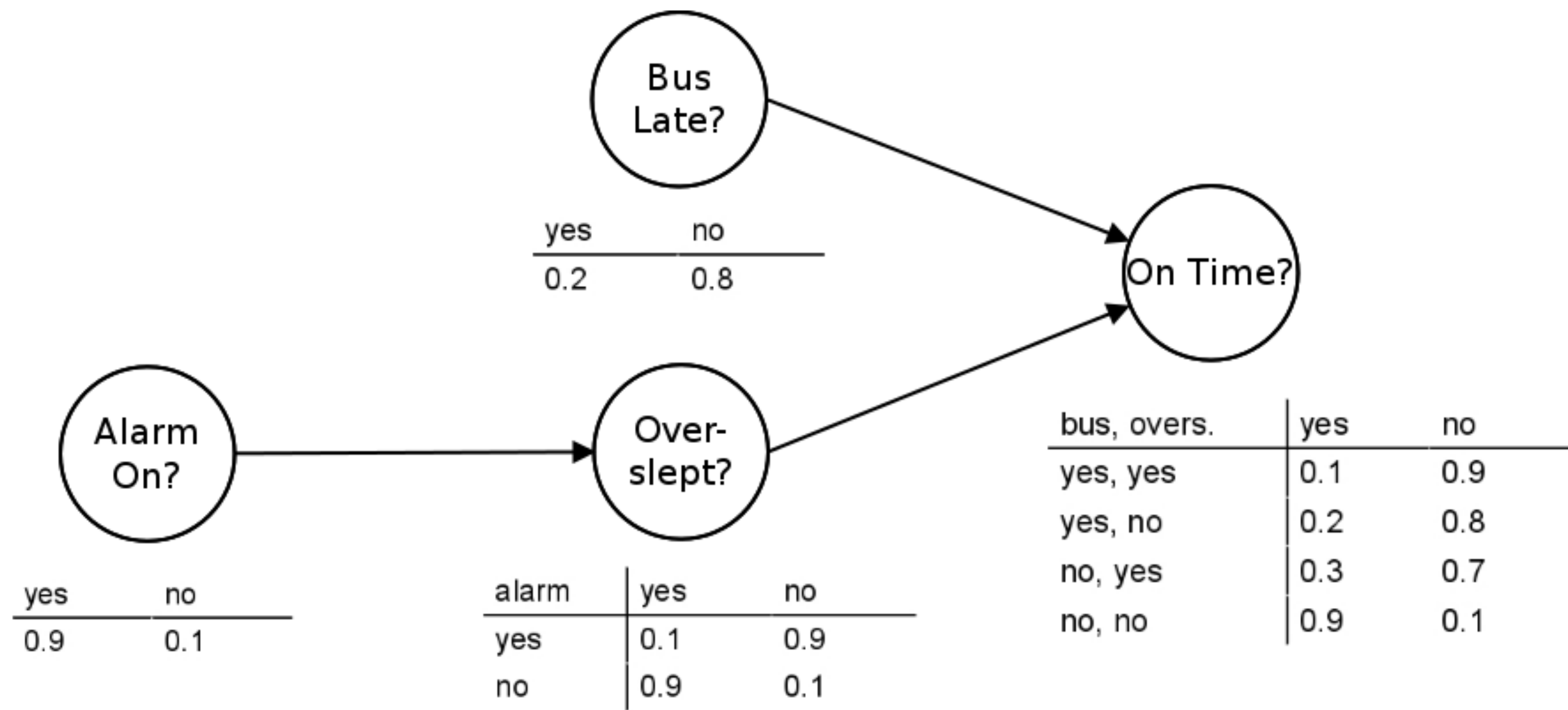


image source: adapted from Fawzi et al. (2021)

Overview of Bayesian Network

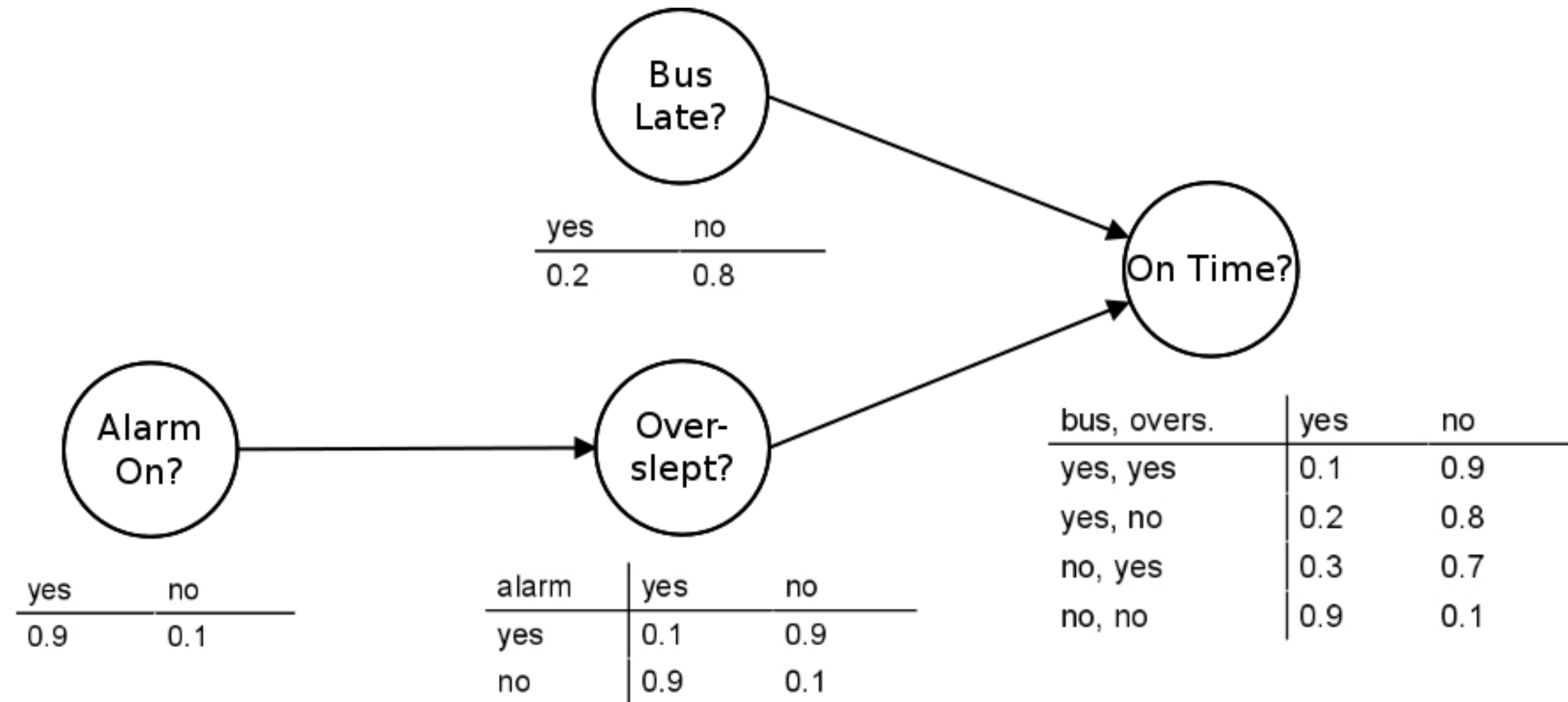
- A Bayesian Network (BN) is a graphical model that represents a set of variables and their probabilistic relationships using a Directed Acyclic Graph (DAG).
- Each node represents a variable; each edge indicates a conditional dependency.
- The graph structure reflects how the probability of one variable depends on others.
- Once the structure is known, we can compute how the probability of any variable changes when we observe others (called inference).
- Benefits:
 - Represents conditional dependencies in DAGs (Pearl, 2009; Koller & Friedman, 2009).
 - Effective when relationships are nonlinear or interdependent.
 - Captures indirect, mediating, and latent effects.

Overview of Bayesian Network



Overview of Bayesian Network

- ex) $P(\text{Overslept} = \text{yes} \mid \text{Alarm} = \text{no}) = 0.9$
- ex) $P(\text{OnTime} = \text{yes} \mid \text{BusLate} = \text{no}, \text{Overslept} = \text{no}) = 0.9$
- $\Rightarrow$ we can construct joint probability distribution of whole network by using these conditional probabilities
- $\Rightarrow$ so that we can inference or reverse inference



Research objective



Explore how individual-level factors
shape media trust

Data description

- Target: Korean adults aged 18 to 79
- Sample: $n = 1,716$; 67% male.
- Quota sampling by gender, age, and region (Jan 26 – Feb 4, 2025)
- Primary variable domain
 - 1. Socio-demographic characteristics
 - 2. Political and social trust
 - 3. Media usage behavior
 - 4. Communication frequency
 - 5. News consumption

Procedure

Data Processing

- Converted all variables to categorical (factors).
- Discretized into 2-3 levels (low/mid/high).

Modeling Procedure

- Hill-Climbing(hc)*, Tabu Search(tabu), and Grow-Shrink(gs) algorithms were used.
 - H-C: A greedy, score-based optimization algorithm for learning the structure of a Bayesian Network (BN). No prior structure needed (purely data-driven)

Measure

- (a) Socio-demographic Characteristics
- Variables: Income, marital status, educational attainment, employment status, subjective social class
 - Income: 12 income brackets → recoded into 3 groups: *low (1-4), middle (5-8), high (9-12)*
 - Marital status: 6 options → grouped into *single, married/cohabiting, others* (widowed/divorced/separated)
 - Education: Ranging from no schooling to Ph.D. → grouped into *low, middle, high*
 - Employment: Binary indicator (1 = employed, 0 = not employed)
 - Subjective social class: 10-rung self-anchoring scale → dichotomized into *low / high* based on the sample mean

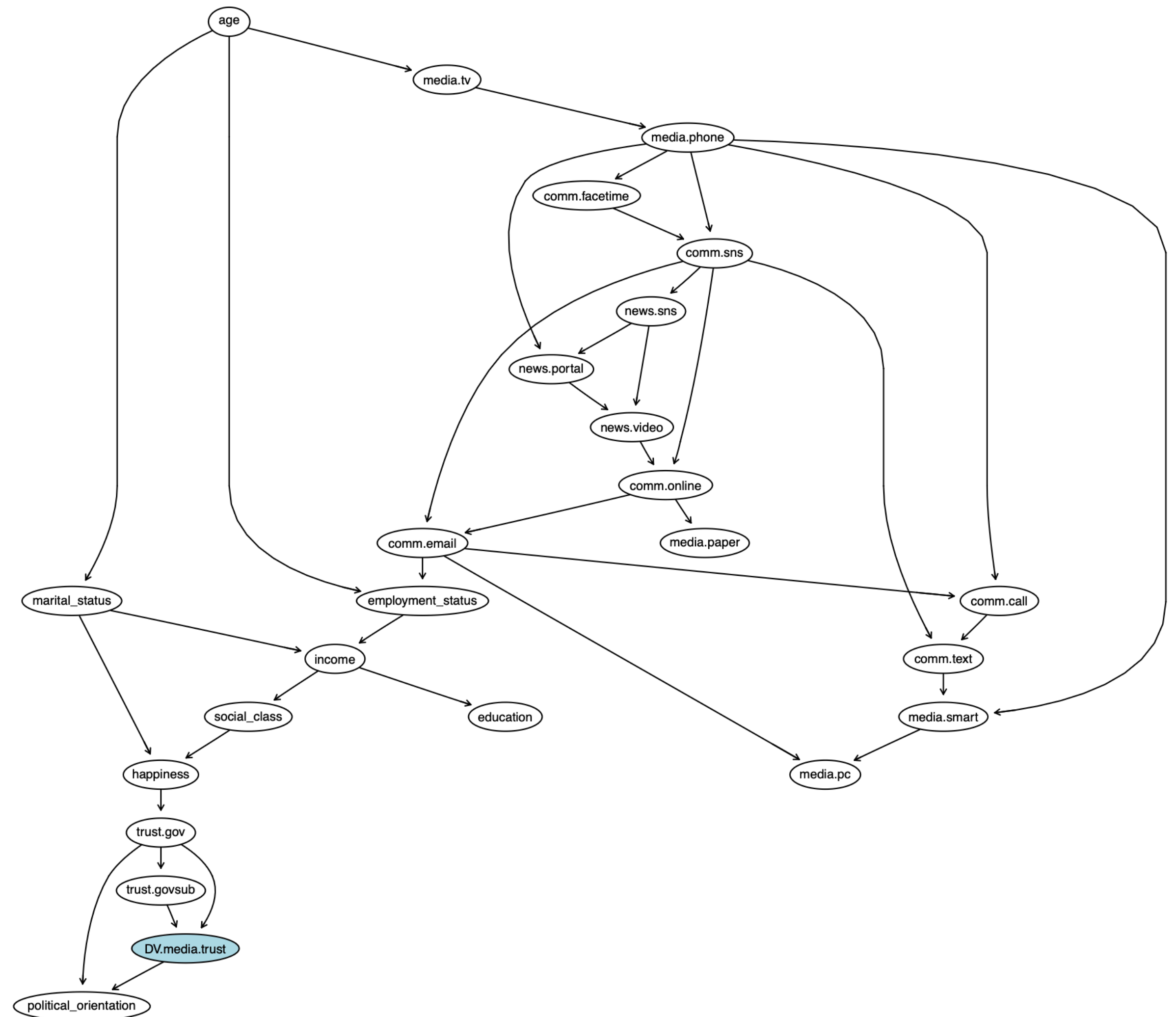
- (b) Political and Social Trust
- Variables: Political orientation, happiness, trust in central and subnational government
- Measurement & Coding:
 - Political orientation: 1 (very progressive) to 7 (very conservative) → recoded into *progressive, moderate, conservative*
 - Happiness: 0–10 scale → dichotomized using the sample mean
 - Government trust (central/subnational): 7-point scale → dichotomized using the sample mean

Measure

- (c) Media Usage Behavior
 - Variables: Frequency of using non-smartphones, smartphones, PCs, television, print media
 - Measurement & Coding:
 - 9-point ordinal scale (1 = never, 9 = multiple times per hour) → recoded into *low* (1-3), *medium* (4-6), *high* (7-9)
- (d) Communication Frequency
 - Variables: Frequency of interpersonal communication via voice calls, text messaging, video calls, SNS, email, online communities
 - Measurement & Coding:
 - 9-point frequency scale → recoded into *low* (1-3), *medium* (4-6), *high* (7-9)

- (e) News Consumption
- Variables: Frequency of consuming news via SNS, video streaming services, internet portals
- Measurement & Coding:
 - 9-point frequency scale → recoded into *low (1–3)*, *medium (4–6)*, *high (7–9)*

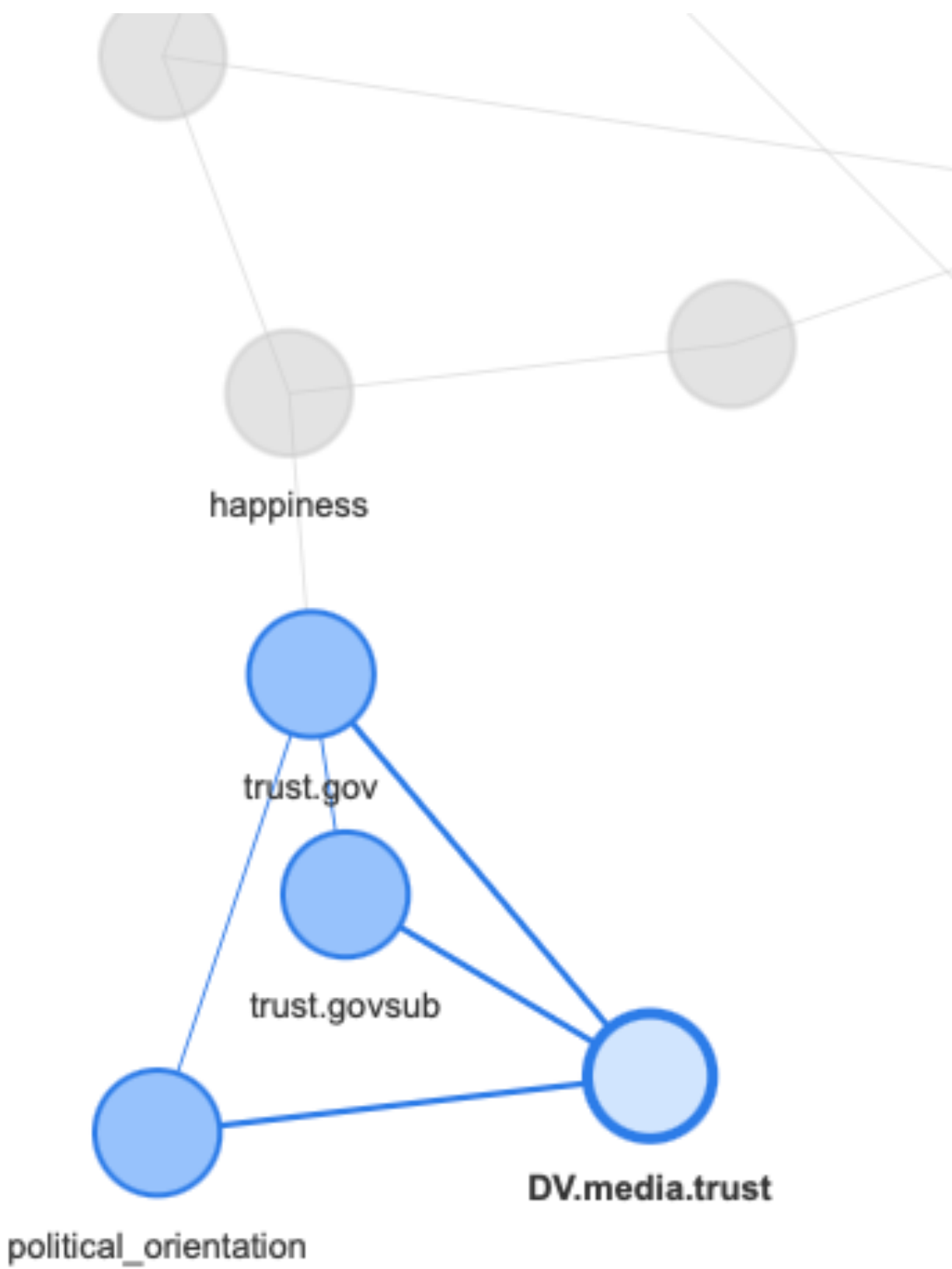
- Marital status, income level, and subjective social class were found to exert significant influence on *subjective happiness*, which in turn affected trust in the *central government*.
- *Central government trust* emerged as a critical antecedent that directly shaped multiple downstream constructs, including *subnational government trust*, *media trust*, and *political orientation*.



- Within the inferred network structure, media trust functions as a child node of both central and subnational government trust, indicating that individuals' trust in governmental institutions directly informs their trust in media sources (=markov blanket).

Table 2. Conditional Probability Table for Media Trust

Central Government Trust	Subnational Government Trust	P(Media Trust=High)	P(Media Trust=Low)
High	High	.686	.314
High	Low	.527	.473
Low	High	.349	.651
Low	Low	.165	.835



Conclusion and implications

- The Bayesian network reveals that trust in central and subnational government forms the backbone of the trust architecture, with media trust operating as a key intermediary that shapes downstream political beliefs.
 - 1. Media trust is shaped by layered dependencies.
 - 2. The positioning of political orientation as a child node of media trust reveals the mediating role that media perceptions play in political identity formation.
- => This reframes media trust not as a singular outcome, but as a strategic variable with implications for democratic functioning, polarization, and information legitimacy.

Limitations

- Cross-sectional data: cannot confirm causality.
- BN is exploratory (not confirmatory) and may overfit the data due to greedy search (e.g., hill-climbing).
- Unmeasured confounders may exist.

Exploring the Probabilistic Causal Structure of Media Trust: A Bayesian Network Analysis

Hye Hyun Ma

Ph.D. Candidate, Dept. of Advertising & Public Relations, Hanyang University

2025. 5. 31.